Data: Samples and Graphs

Name: Date:

Learning goal: I can choose a fair sample and read and question stacked bar graphs and misleading graphs.

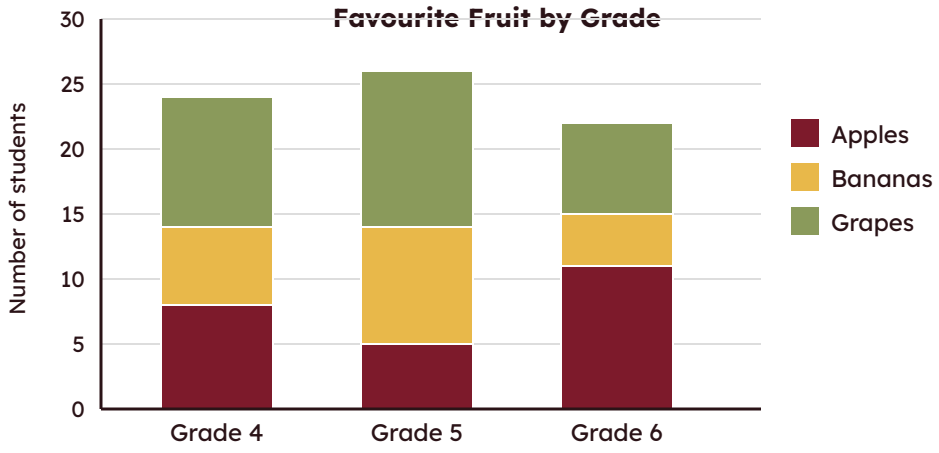
1 A student wants to know the **favourite sport of all students** at the school. Is each sample fair (representative)? Write **yes** or **no** and explain.

a) Ask only the basketball team.

b) Ask every 5th student on each class list, from Kindergarten to Grade 8.

c) Ask only the Grade 1 class.

2 Use the stacked bar graph.



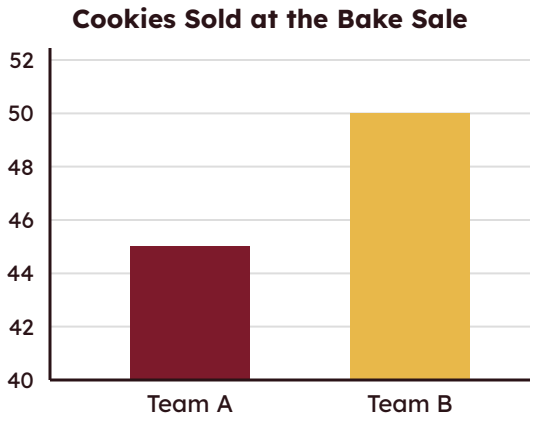
a) How many Grade 5 students were surveyed?

b) Which fruit was most popular in Grade 6?

c) How many students in all chose grapes?

d) Which grade had the most students surveyed?

3 Look at this graph.



Team B's bar looks about **twice** as tall as Team A's. Did Team B really sell twice as many cookies? Why is this graph misleading?

4 Students named their pets: *dog, cat, dog, fish, none, dog, cat, dog, bird, cat, dog, none, fish, dog, cat.* Make a frequency table.

Pet	Tally	Frequency
Dog		
Cat		
Fish		
Bird		
None		

TIP FOR FAMILIES

Look for graphs in flyers, the news or sports pages and ask: What is this graph telling us? Does the scale start at zero? Who was asked? These questions build strong data sense.

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Key

Answers

Try the handout first, then check together. Where it says “answers vary,” any correct example counts.

1. a) No, basketball players will likely choose basketball. b) Yes, every grade is included and students are chosen fairly. c) No, one grade does not represent the whole school.
2. a) 26 (5 + 9 + 12) b) Apples (11) c) 29 (10 + 12 + 7) d) Grade 5
3. No. Team A sold 45 and Team B sold 50, only 5 more. The scale starts at 40 instead of 0, so the difference looks much bigger than it is.
4. Dog 6, Cat 4, Fish 2, Bird 1, None 2 (total 15)